

Fever of Unknown Origin, Toxic Shock, Catheter-Related Sepsis, Neutropenic Fever, and Infectious Mononucleosis

Chapter 8F

Medvale Internal Medicine | Part VIII: Infectious Diseases

This chapter is a focused infection-syndrome bridge. It covers persistent fever without a clear source, toxin-mediated shock, line-associated bloodstream infection, fever in neutropenia, common organism patterns, and mononucleosis-like syndromes. The emphasis is on converting fever into a structured differential, an initial workup, empiric treatment, and follow-up plan.

Why this matters clinically

Fever is a physiologic signal rather than a diagnosis. The same measured temperature can represent self-limited viral illness, occult abscess, endocarditis, malignancy, vasculitis, drug fever, hyperthermia, or life-threatening sepsis.

Infectious disease reasoning therefore depends on context: host immune status, devices, prosthetic material, exposures, localizing symptoms, hemodynamic stability, and time course.

The syndromes in this chapter are dangerous because early clues may be incomplete. Toxic shock syndrome can have negative blood cultures because illness is toxin-mediated. Catheter-related sepsis can occur with a normal-looking insertion site. Neutropenic fever may have no pus, no focal infiltrate, and no leukocytosis. Infectious mononucleosis can mimic streptococcal pharyngitis, acute HIV, CMV, or leukemia. Fever of unknown origin requires diagnostic patience rather than random broad therapy in stable patients.

Core rule: do not delay treatment when fever occurs with shock, neutropenia, meningitis concern, severe immunosuppression, toxic appearance, necrotizing soft tissue concern, or unstable device-associated infection. Draw cultures quickly when feasible, then treat and pursue source control.

Core illness scripts

Syndrome patterns

Fever of unknown origin

Persistent documented fever without diagnosis after appropriate initial evaluation. The workup relies on serial history, repeated examination, targeted cultures, inflammatory markers, imaging, and tissue diagnosis when indicated.

Toxic shock syndrome

Abrupt fever, hypotension, diffuse erythematous rash, vomiting or diarrhea, myalgias, mucous membrane hyperemia, and multi-organ injury after toxin production by *Staphylococcus aureus* or group A *Streptococcus*.

Catheter-related bloodstream infection

Central venous catheter, ICU stay, dialysis catheter, TPN, or prolonged line dwell time with fever or sepsis and no other source. The line site may look normal.

Neutropenic fever

Fever in a host with ANC below 500 cells/microliter or expected to fall below that level. Fever may be the only manifestation because neutrophil-dependent inflammation is absent.

Infectious mononucleosis

Fever, fatigue, pharyngitis, posterior cervical lymphadenopathy, atypical lymphocytosis, and splenomegaly, most often from EBV. Acute HIV, CMV, toxoplasmosis, and streptococcal pharyngitis are important mimics.

Fever physiology and fever versus hyperthermia

Fever is mediated by a regulated increase in hypothalamic temperature set point. Microbial products and endogenous cytokines such as interleukin-1, interleukin-6, tumor necrosis factor, and interferons increase prostaglandin E2 signaling near the hypothalamus. The patient feels cold and shivers while the body rises toward the new set point. Rigors are intense shaking chills; they suggest abrupt cytokine signaling and can occur with bacteremia, malaria, pyelonephritis, cholangitis, pneumonia, or severe viral syndromes, but they are not organism-specific.

Hyperthermia is different. In hyperthermia the body temperature rises because heat production or environmental exposure overwhelms heat loss, without a hypothalamic set-point increase. Heat stroke, serotonin syndrome, neuroleptic malignant syndrome, and malignant hyperthermia are examples. Antipyretics treat prostaglandin-mediated fever but are not sufficient for hyperthermia; external cooling and syndrome-specific therapy are required.

Fever language

Chills

Subjective cold sensation during a rising set point; common but nonspecific.

Rigors

Severe shaking chills; think bacteremia, malaria, cholangitis, pyelonephritis, pneumonia, or severe systemic infection.

Fever

Regulated set-point increase; often improves with antipyretics, but response does not prove benign disease.

Hyperthermia

Unregulated heat excess; antipyretics alone are inadequate.

Hypothermia in infection

Can occur in severe sepsis, older adults, exposure, adrenal disease, thyroid disease, or shock; may signal severe physiology.

Fever of unknown origin

Fever of unknown origin is persistent documented fever that remains unexplained after appropriate evaluation. Classic definitions required temperature above 38.3 C on multiple occasions, duration longer than 3 weeks, and no diagnosis after a defined inpatient workup. Modern practice is less rigid because outpatient imaging and testing are available earlier. The key idea is that FUO is a term of exclusion, not a final diagnosis.

The major categories are infection, malignancy, noninfectious inflammatory disease, and miscellaneous causes. Infection includes tuberculosis, endocarditis, occult abscess, complicated UTI, sinusitis, osteomyelitis, infected devices, HIV, EBV, CMV, malaria, and fungal disease. Malignancy includes lymphoma, leukemia, renal cell carcinoma, hepatocellular carcinoma, and metastatic cancer. Inflammatory disease includes giant cell arteritis, adult-onset Still disease, SLE, vasculitis, sarcoidosis, inflammatory bowel disease, and polymyalgia rheumatica. Miscellaneous causes include drug fever, pulmonary embolism, thyroiditis, factitious fever, hemolytic anemia, gout, and familial periodic fever syndromes.

FUO workup pattern

Confirm fever

Document real temperatures and time course. Distinguish fever from hyperthermia, measurement error, and subjective warmth.

History

Review medications, travel, animals, TB risk, sexual exposure, occupation, devices, dental work, transfusions, immune status, and family history.

Repeat physical exam

Look again for murmur, rash, lymphadenopathy, temporal artery tenderness, arthritis, spinal tenderness, abdominal mass, hepatosplenomegaly, and fundoscopic lesions.

Baseline labs

CBC with differential, CMP, liver tests, ESR/CRP, urinalysis, HIV testing, and blood cultures when indicated. Cytopenias can suggest marrow disease, viral infection, HLH, hypersplenism, or malignancy.

Imaging

Use chest radiograph, ultrasound, CT, MRI, or PET/CT based on syndrome and pretest probability. Look for abscess, malignancy, lymphadenopathy, vasculitis, osteomyelitis, or infected prosthetic material.

Tissue diagnosis

Biopsy accessible lymph nodes, temporal artery, bone marrow, liver, skin lesion, or mass when indicated. Tissue often solves persistent FUO when blood tests do not.

FUO trap: in stable patients, empiric antibiotics can sterilize cultures and delay diagnosis. Exceptions include instability, neutropenia, meningitis, unstable endocarditis concern, severe TB in high-risk hosts, or another syndrome where delay is dangerous.

Sepsis and source control

Sepsis is life-threatening organ dysfunction caused by a dysregulated host response to infection. Bedside concern rises when fever or suspected infection is accompanied by hypotension, altered mental status, hypoxemia, oliguria, rising creatinine, liver injury, thrombocytopenia, coagulopathy, or elevated lactate. Septic shock adds circulatory and cellular-metabolic failure, often requiring vasopressors after fluids and showing elevated lactate.

Treatment is not just antibiotics. The sequence is recognition, cultures when they do not delay therapy, empiric antimicrobials matched to syndrome and host risk, hemodynamic resuscitation, source identification, and source control. Source control means draining pus, removing infected lines, debriding necrotic tissue, relieving obstruction, or operating on perforation. Antibiotics alone do not reliably cure undrained pus or devitalized tissue.

Sepsis decision points

Possible sepsis

Fever or hypothermia, tachycardia, tachypnea, leukocytosis or leukopenia, suspected infection. Assess organ dysfunction and obtain cultures/lactate when appropriate.

Septic shock physiology

Persistent hypotension, altered mentation, oliguria, elevated lactate, cool or mottled extremities. Give fluids when hypoperfused and use vasopressors if needed.

Source control need

Abscess, infected catheter, obstructed pyelonephritis, obstructed cholangitis, necrotizing infection, perforated bowel, infected prosthetic material. Call procedural or surgical teams early.

De-escalation moment

When cultures and susceptibilities return and the patient stabilizes, narrow therapy, stop unnecessary MRSA or anaerobic or antifungal coverage, and define duration.

Toxic shock syndrome

Toxic shock syndrome is an acute toxin-mediated illness caused most often by *Staphylococcus aureus* or group A *Streptococcus*. Superantigen toxins cause widespread T-cell activation, cytokine release, capillary leak, hypotension, rash, gastrointestinal symptoms, myalgias, and organ dysfunction. Sources include tampons, wounds, burns, nasal packing, postpartum or postoperative sites, and necrotizing soft tissue infection.

Classic findings include abrupt fever, diffuse erythematous rash, hypotension, vomiting or diarrhea, mucous membrane hyperemia, severe myalgias, renal injury, hepatic injury, thrombocytopenia, altered mental status, and later desquamation of palms and soles. The rash can be subtle or delayed. Severe pain out of proportion to skin findings should trigger concern for necrotizing infection with toxin-mediated physiology.

Staphylococcal versus streptococcal toxic shock

Staphylococcal TSS

Classically associated with tampon use, nasal packing, wounds, burns, or postoperative sites. Blood cultures are often negative because toxin production rather than bloodstream invasion drives illness.

Streptococcal TSS

Often associated with invasive group A strep infection, cellulitis, myositis, pneumonia, postpartum infection, or necrotizing fasciitis. Blood or tissue cultures are more often positive.

Treatment overlap

Resuscitate shock, remove the toxin source, cover staphylococci and streptococci, add toxin-suppressing therapy such as clindamycin when suspected, and obtain surgical evaluation for deep tissue disease.

Do not anchor on menstruation

Menstrual TSS is classic, but nonmenstrual TSS occurs with wounds, burns, nasal packing, postpartum infections, surgical sites, and invasive group A streptococcal disease.

Catheter-related bloodstream infection and catheter-related sepsis

Central venous catheters are a major source of healthcare-associated bloodstream infection. Risk increases with emergent placement, femoral access, prolonged dwell time, total parenteral nutrition, hemodialysis catheters, neutropenia, poor line care, and heavy catheter manipulation. Common pathogens include coagulase-negative staphylococci such as *Staphylococcus epidermidis*, *Staphylococcus aureus*, *Enterococcus*, gram-negative bacilli, and *Candida*.

Only a minority of patients have obvious insertion-site inflammation, so a normal-appearing line does not exclude line infection. Diagnosis is supported by paired blood cultures from catheter and peripheral vein, differential time to positivity, and catheter tip culture after removal. Management depends on pathogen, severity, catheter type, alternative access, and evidence of tunnel infection, port pocket infection, persistent bacteremia, endocarditis, septic thrombosis, or metastatic infection.

Line infection interpretation

Fever with central line and no other source

Obtain peripheral and catheter blood cultures; start empiric therapy when clinically ill.

Purulence, tunnel tenderness, or port pocket infection

Remove catheter and culture the tip; cover likely gram-positive organisms and broaden by severity.

S. aureus, *Candida*, *Pseudomonas*, persistent bacteremia

High-risk pathogens usually favor catheter removal and evaluation for metastatic infection.

One positive culture with skin flora

May be contamination, but multiple positive cultures, prosthetic material, and clinical instability shift the interpretation toward true infection.

Persistent bacteremia after line removal

Evaluate for endocarditis, septic thrombosis, osteomyelitis, abscess, or retained infected material.

Neutropenic fever

Neutropenic fever is an emergency because the absence of neutrophils removes many localizing signs. Cellulitis may lack pus, pneumonia may initially lack an infiltrate, meningitis may lack marked CSF pleocytosis, and abdominal infection may have muted peritoneal findings. A single fever in severe neutropenia can be the first sign of life-threatening bacteremia.

Absolute neutrophil count is calculated as WBC multiplied by the percentage of segmented neutrophils plus bands. Risk rises when ANC is below 500 cells/microliter or expected to fall below 500. Profound neutropenia below 100 and prolonged duration beyond 7 days increase risk for invasive bacterial and fungal infection. Common infection sources include gut translocation, catheter-related bloodstream infection, pneumonia, skin and soft tissue infection, perirectal infection, urinary infection, mucositis, sinusitis, and invasive fungal infection after persistent fever.

Risk stratification and therapy

Lower-risk pattern

ANC rising, expected short duration, hemodynamically stable, no organ dysfunction, no major comorbidities, no pneumonia or abdominal process, reliable follow-up.

Higher-risk pattern

ANC below 500, especially below 100; prolonged neutropenia; hypotension; hypoxia; altered mental status; pneumonia; abdominal pain; renal or hepatic dysfunction; transplant; high-dose steroids; uncontrolled cancer.

Initial therapy

High-risk patients need immediate IV anti-pseudomonal therapy after rapid cultures, commonly cefepime, piperacillin-tazobactam, or meropenem based on local resistance and severity.

When to add vancomycin

Add for catheter infection, skin or soft tissue infection, pneumonia, hemodynamic instability, known MRSA colonization, or resistant gram-positive concern. It is not routine for every neutropenic fever.

Persistent fever

After several days of broad antibacterial therapy with ongoing high-risk neutropenia, reassess for occult source and consider invasive fungal infection and antifungal therapy.

IDSA and ASCO/IDSA guidance treats fever and neutropenia in cancer patients as a condition requiring prompt empiric antibacterial therapy. Outpatient treatment is reserved for carefully selected low-risk patients with reliable follow-up.

Common fever sources by syndrome

Organism maps for empiric reasoning

Community pneumonia

S. pneumoniae, *H. influenzae*, atypicals, respiratory viruses; aspiration risk adds oral anaerobes.

Hospital or ventilator pneumonia

Gram-negative rods including *Pseudomonas* and *S. aureus* including MRSA; prior IV antibiotics and local antibiogram matter.

UTI and pyelonephritis

E. coli, *Klebsiella*, *Proteus*, *Enterococcus*, *Staph saprophyticus* in young women; obstruction, catheter, male sex, pregnancy, diabetes, or stones make it complicated.

Skin and soft tissue infection

Strep pyogenes and *S. aureus* predominate; diabetic foot and bite wounds are often polymicrobial; necrotizing infection requires surgery.

Endocarditis

Viridans streptococci, *S. aureus*, enterococci, HACEK organisms, and fungi in selected hosts; IV drug use and prosthetic valves change the likely organisms.

Intra-abdominal infection

Enterobacterales, anaerobes such as *Bacteroides*, and enterococci; source control is often as important as antibiotics.

Catheter-related bloodstream infection

Coagulase-negative staphylococci, *S. aureus*, *Candida*, and gram-negative rods; the line site may look normal.

Neutropenic fever

Gut translocation gram-negative rods, catheter organisms, pneumonia pathogens, and later Candida or molds with prolonged fever and broad antibiotics.

Infectious mononucleosis and mononucleosis-like syndromes

Infectious mononucleosis is most commonly caused by Epstein-Barr virus. CMV can cause a similar heterophile-negative syndrome. Transmission is usually through saliva, and the incubation period is several weeks. The classic triad is fever, pharyngitis, and lymphadenopathy, especially posterior cervical lymphadenopathy. Fatigue can persist for weeks to months. Splenomegaly is common enough that contact sports and strenuous activity must be restricted because splenic rupture, though rare, can be catastrophic.

The differential includes group A streptococcal pharyngitis, acute HIV, CMV, toxoplasmosis, viral hepatitis, leukemia or lymphoma, gonococcal pharyngitis, and diphtheria in appropriate contexts. Acute HIV deserves special attention because it may produce fever, pharyngitis, rash, lymphadenopathy, mucocutaneous ulcers, diarrhea, or aseptic meningitis. Missing acute HIV delays treatment and increases transmission risk.

Mononucleosis comparison

EBV mononucleosis

Adolescent or young adult; fever, exudative pharyngitis, posterior cervical nodes, splenomegaly, atypical lymphocytosis, mild transaminitis, and heterophile positivity after the earliest phase.

CMV mononucleosis

Heterophile-negative mono-like illness, often less prominent pharyngitis; important in pregnancy, transplant, or transfusion contexts.

Acute HIV

Fever, sore throat, rash, lymphadenopathy, diarrhea, mucocutaneous ulcers, or aseptic meningitis after recent exposure. Test with HIV antigen/antibody and HIV RNA when very early infection is possible.

Group A strep pharyngitis

Tender anterior cervical nodes, fever, tonsillar exudates, and absence of cough support the diagnosis; confirm with rapid antigen testing or culture when indicated.

Counseling

Avoid contact sports and heavy exertion until splenic risk has resolved. Steroids are reserved for airway compromise or severe immune complications, not routine fatigue.

Integrated diagnostic algorithms

Algorithm 1: Stable fever without a source

Confirm

Is this true fever, hyperthermia, or measurement error? Repeat measured temperature and manage hyperthermia urgently if present.

Danger screen

Look for shock, hypoxia, altered mentation, meningismus, neutropenia, transplant, severe pain, or toxic appearance. If present, treat as an emergency syndrome.

Localize

Ask about cough, dysuria, abdominal pain, rash, line, joint, headache, travel, tick exposure, animal exposure, and sexual exposure. Order focused tests and empiric therapy by syndrome.

Host risk

Devices, prosthetic valves or joints, immune defects, recent antibiotics, and hospitalization broaden the differential to resistant organisms, fungi, device infection, and opportunistic infection.

Persistent fever

If fever continues for weeks or after repeated nondiagnostic visits, shift to the FUO pathway: repeat history/exam, inflammatory markers, cultures, imaging, and tissue diagnosis when indicated.

Algorithm 2: Fever plus rash plus hypotension

Diffuse rash, vomiting, mucosal hyperemia

Think toxic shock syndrome. Resuscitate, remove source, give anti-staph/strep coverage, and add toxin-suppressing therapy.

Pain out of proportion, bullae, crepitus, skin anesthesia

Think necrotizing soft tissue infection. Call surgery immediately and give broad antibiotics.

Petechiae or purpura with meningismus or shock

Think meningococcemia or meningitis. Use droplet precautions and prompt ceftriaxone-based therapy.

Tick exposure with rash from wrists/ankles to trunk

Think Rocky Mountain spotted fever. Give doxycycline immediately; do not wait for serology.

Sore throat, rash, lymphadenopathy after exposure risk

Think acute HIV or viral syndrome. Order HIV antigen/antibody and RNA when indicated.

Algorithm 3: Fever with a central line

Step 1

Assess instability. If shock, hypoxia, altered mentation, or neutropenia is present, treat as sepsis while obtaining cultures rapidly.

Step 2

Draw paired cultures: peripheral blood and catheter lumen cultures when available.

Step 3

Inspect insertion site, tunnel, and port pocket, but do not rule out infection because the site looks normal.

Step 4

Start empiric therapy based on severity, MRSA risk, gram-negative risk, Candida risk, and local antibiogram.

Step 5

Remove catheter for *S. aureus*, Candida, Pseudomonas, tunnel/pocket infection, severe sepsis, persistent bacteremia, or lack of another explanation when clinically indicated.

Step 6

Repeat cultures to document clearance and evaluate for endocarditis, septic thrombosis, osteomyelitis, or metastatic foci if bacteremia persists.

Practice cases

Integrated cases

Case 1

A patient with fever, vomiting, hypotension, diffuse erythematous rash, tampon use, and negative blood cultures most likely has staphylococcal toxic shock syndrome. Treat with resuscitation, source removal, anti-staph coverage, and toxin suppression.

Case 2

A chemotherapy patient with ANC 80 and temperature 38.4 C has high-risk neutropenic fever even if the chest radiograph is normal. Obtain cultures quickly and start IV anti-pseudomonal beta-lactam therapy.

Case 3

An ICU patient with a femoral central line, fever, and hypotension but no insertion-site redness may still have catheter-related sepsis. Draw paired cultures and consider catheter removal based on severity and pathogen.

Case 4

A college student with fever, fatigue, exudative pharyngitis, posterior cervical nodes, atypical lymphocytes, and splenomegaly most likely has EBV mononucleosis. Provide supportive care and activity restriction.

Case 5

A patient with 4 weeks of fever, weight loss, high ESR, new headache, and jaw claudication may have giant cell arteritis as a noninfectious FEO cause. Do not wait for cultures to treat suspected vision-threatening disease.

Case 6

A diabetic patient with fever, hypotension, severe calf pain, and minimal erythema has possible necrotizing fasciitis with toxic shock physiology. Surgery is the treatment priority.

Common pitfalls

- Treating FEO as a reason for random broad antibiotics in a stable patient rather than using a staged diagnostic evaluation.
- Excluding catheter-related infection because the insertion site is not red or purulent.
- Waiting for neutropenic fever to localize before treating.
- Forgetting source removal and toxin suppression in toxic shock syndrome.
- Missing acute HIV in a mononucleosis-like syndrome, especially when heterophile testing is negative.
- Assuming fever always means infection; malignancy, vasculitis, thrombosis, drug fever, thyroiditis, and factitious fever can mimic infection.
- Overinterpreting one positive blood culture with skin flora without considering number of bottles, clinical context, and device or prosthetic material.
- Failing to counsel mononucleosis patients about splenic rupture risk and activity restriction.

Graph-RAG extraction map

Nodes and edges for future extraction

Fever of unknown origin

Class: diagnostic framework. Facts: persistent fever without diagnosis after appropriate evaluation; categories include infection, malignancy, inflammatory disease, and miscellaneous causes. Edges: FEO to occult abscess, endocarditis, lymphoma, vasculitis, drug fever.

Toxic shock syndrome

Class: toxin-mediated shock. Facts: abrupt fever, hypotension, rash, GI symptoms, organ injury; staph or GAS toxins. Edges: TSS to source removal, clindamycin/toxin suppression, necrotizing fasciitis.

Catheter-related bloodstream infection

Class: device-associated infection. Facts: line may look normal; coagulase-negative staph, *S. aureus*, *Candida*, and gram-negative rods. Edges: CRBSI to paired cultures, catheter removal, endocarditis evaluation.

Neutropenic fever

Class: immunocompromised emergency. Facts: ANC below 500 or expected decline; signs muted; anti-pseudomonal therapy urgent. Edges: neutropenic fever to *Pseudomonas* coverage, fungal infection, catheter infection.

Infectious mononucleosis

Class: viral syndrome. Facts: EBV; fever, pharyngitis, posterior cervical nodes, splenomegaly, atypical lymphocytes. Edges: mono to splenic precautions, acute HIV mimic, heterophile test.

Sepsis

Class: critical illness physiology. Facts: organ dysfunction from infection; source control and timely antimicrobials are central. Edges: sepsis to lactate, source control, empiric antibiotics, vasopressors.

Quick review questions

Why can toxic shock syndrome have negative blood cultures?

Answer: Systemic illness is often driven by exotoxin rather than sustained bacteremia, especially in staphylococcal TSS.

What is the first antibiotic priority in high-risk neutropenic fever?

Answer: Prompt empiric anti-pseudomonal coverage such as cefepime, piperacillin-tazobactam, or meropenem based on local resistance and patient factors.

Why can catheter-related sepsis be missed on exam?

Answer: Many patients have no erythema or purulence at the insertion site, so paired blood cultures and clinical suspicion matter.

What findings support EBV mononucleosis over routine bacterial pharyngitis?

Answer: Posterior cervical lymphadenopathy, atypical lymphocytosis, splenomegaly, prolonged fatigue, and positive heterophile or EBV-specific testing.

Name noninfectious causes of FUO.

Answer: Lymphoma, leukemia, vasculitis, giant cell arteritis, SLE, adult-onset Still disease, drug fever, pulmonary embolism, thyroiditis, sarcoidosis, and factitious fever.

Why does source control matter in sepsis?

Answer: Undrained abscesses, infected catheters, obstructed systems, necrotic tissue, and foreign bodies can perpetuate infection despite antibiotics.

References and source note

This chapter is original Medvale educational content. The uploaded infection5.pdf was used as a coverage guide for fever of unknown origin, toxic shock syndrome, catheter-related sepsis, neutropenic fever, common organism patterns, and infectious mononucleosis.

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- CDC. Clinical guidance for streptococcal toxic shock syndrome and CDC toxic shock syndrome surveillance criteria.
- CDC infection-control isolation precautions and CDC/NIH resources for EBV and mononucleosis-like illness where applicable.